

Cary Jones

Python Developer | Embedded Systems, Device Integration & Hardware-Software Verification

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SUMMARY

Python Developer with 15+ years of production software experience and a foundational **5-year background in embedded medical device systems** at Hospira, verifying software for **Symbiq** and **Plum A+** infusion pumps. Experienced collaborating with firmware and hardware engineers to test, debug, and integrate software with embedded platforms — including **CAN bus** and serial device communication over wired/wireless hospital networks. Builds clean Python applications, automation scripts, and test tooling; documents designs and test plans; troubleshoots complex hardware-software integration issues in regulated, high-reliability environments.

TECHNICAL SKILLS

- **Python:** FastAPI, Django, Flask, Asyncio, pytest, automation scripts, REST clients, SDLC ownership from design through deployment
- **Embedded & Hardware Integration:** Medical infusion devices (Symbiq, Plum A+), embedded systems verification, CAN bus, serial/TCP/IP host-to-device communication, firmware upgrade/recovery, wired & wireless LAN integration
- **Testing & Debugging:** Requirements-based testing, smoke/regression testing, formal verification (FDA-regulated workflows), defect tracking, embedded environment troubleshooting
- **Collaboration:** Cross-functional work with firmware developers, hardware engineers, field service, and verification teams
- **Cloud & Backend:** AWS, Docker, microservices, PostgreSQL, CI/CD — modern Python platform development complementing embedded integration experience
- **Tools & Environments:** Linux, Windows server environments, Telnet/device communication tooling, network security test architectures (EAP, FreeRadius)

KEY IMPACT

- **5 years** verifying embedded infusion pump software (Symbiq, Plum A+) — delivered **5 formal verification run-throughs** for first-lot-to-stock ahead of schedule.
- Built host-to-device Telnet automation tooling and recovery scripts restoring pump software/configuration after firmware failures.
- Architected enterprise wireless security test environment for embedded device verification (EAP, Cisco ACS, FreeRadius).

- Led software upgrade of **75 hospital infusion devices** over two days for a sales/marketing conference deployment.
- 15+ years subsequent Python engineering — microservices, AWS pipelines (**2.5M+ daily events**), and production systems at **99.99% uptime**.

PROFESSIONAL EXPERIENCE

Quality Assurance & Network Integration Engineer

February 2005 – December 2009 | Hospira, Inc. — San Diego, CA

- Verified and validated embedded software for **Symbiq** and **Plum A+** hospital infusion pumps — requirements-based testing, weekly smoke tests, formal verification, and defect documentation in regulated medical device workflows.
 - Collaborated daily with **firmware and hardware engineers** to test device behavior, debug software upgrades, and resolve integration issues between pumps, drug library servers, and hospital networks.
 - Supported embedded systems over wired and wireless LANs; developed recovery scripts and tools to restore device software and factory configuration after firmware upgrade failures.
 - Worked with **CAN bus** and embedded communication architectures; built a .NET Telnet client automating host-to-device communication for verification and field-support use cases.
 - Served as verification team technical lead for infusion device and PC networking issues; installed and validated client/server drug library software for networked pump deployments.
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Principal Software Engineering Consultant / Technical Lead

Jan 2023 – Present | Los Angeles, CA / Remote

CGJSoftware, Inc.

- Design, develop, test, and deploy Python backend applications and automation tooling on AWS; write clean, documented, maintainable code with unit testing and CI/CD.
 - Debug production issues, optimize performance, and collaborate with cross-functional teams on integration and delivery.
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Senior Software Engineer

October 2021 – November 2022 | Disney Corporation, Inc. (Remote)

- Built and tested high-reliability backend modules; achieved **99.99% uptime** on mission-critical enterprise systems.
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Software Developer in Test

October 2015 – May 2016 | Rackspace, Inc. — Austin, TX

- Developed Python REST test clients executing **1,000+ automated test cases per hour**; built dynamic test generators for edge-case discovery.
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Python Programmer

May 2016 – July 2016 | Dell, Inc. — Austin, TX

- Refactored Python automated test frameworks for enterprise server hardware testing; improved long-running test stability by **80%**.
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Field Wireman & Communications Specialist

September 1995 – September 1999 | United States Marine Corps

- Deployed and maintained mission-critical communications systems under strict uptime and reliability requirements.

EDUCATION

Associate of Science, Computer Electronics Engineering Technology

2010 – 2013 | ITT Technical Institute, San Diego, CA (GPA 3.67)

- Equivalent practical experience: 15+ years professional software engineering including 5 years embedded medical device verification